Data drive analytic/ consulting capstone project Executive Summary

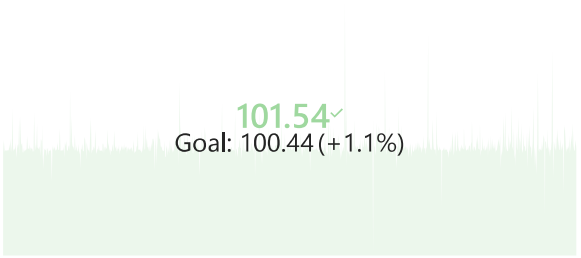
Insight: Total sales are 2.30M and total profit is 286.40K, so the overall profit margin is only 12.47%. Profit has grown every year (2014 to 2017), so the real problem is low margin, not falling profit. Q4 is the peak sales period.

Total Sales	Total Profit	Profit Margin %	Total Orders
2.30M	286.40K	12.47%	5K

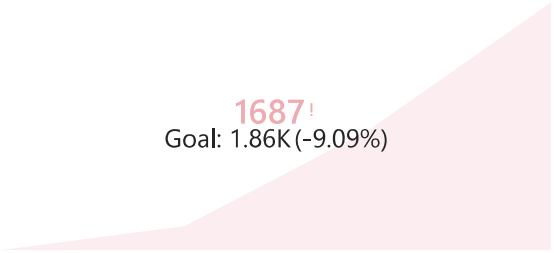
Total Sales and Sales Target (Dynamic)
by Year



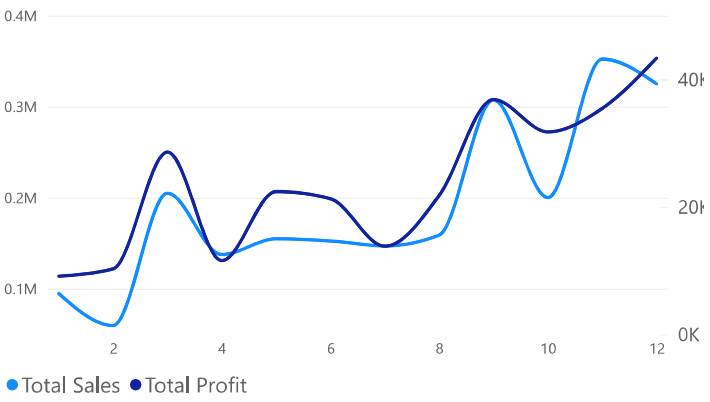
Total Profit and Profit Target
by Order Date

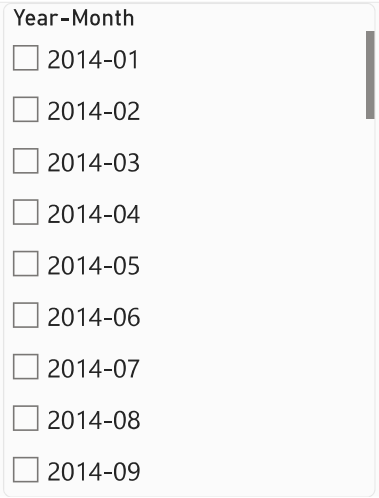


Total Orders and Orders Target
by Year

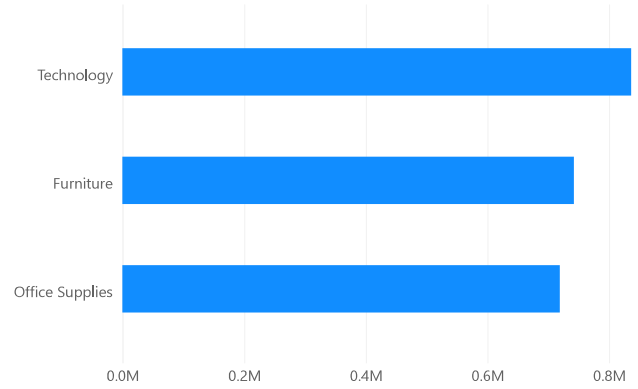


Sales & Profit Trend By Month
Total Sales and Total Profit by Month





Sales by Category
Total Sales by Category



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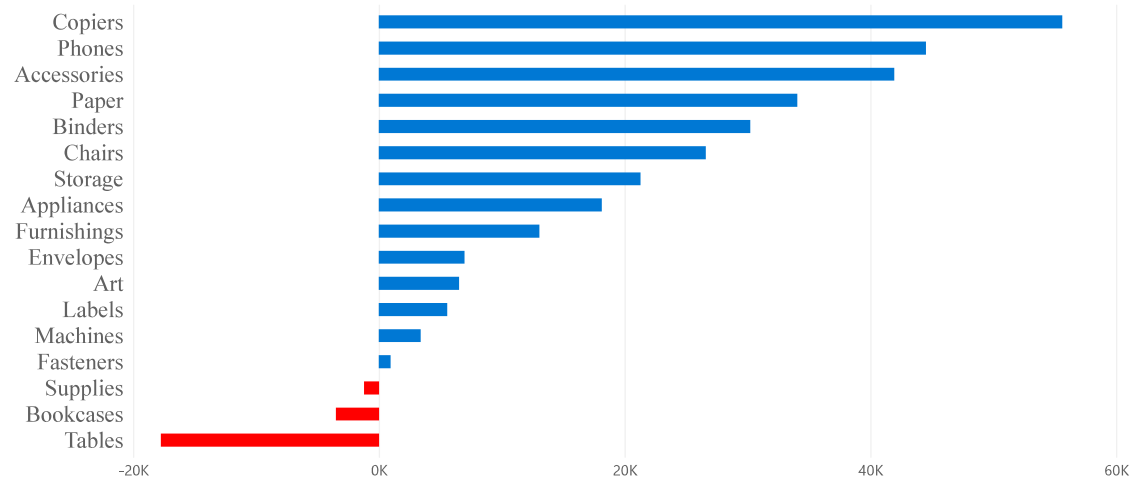
Category & Product Analysis

Insight:

Copiers and Phones are the most profitable sub-categories. Technology drives the maximum sales. Tables, Bookcases and Supplies are loss-making overall. Tables and Bookcases lose money mainly because of high discounts, while Supplies has a low discount, so its loss needs a cost and pricing review.

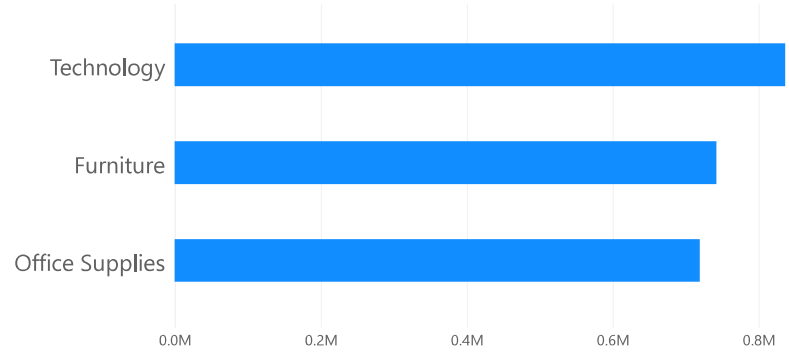
Profit by Sub-Category

Total Profit by Sub-Category



Sales by Category

Total Sales by Category



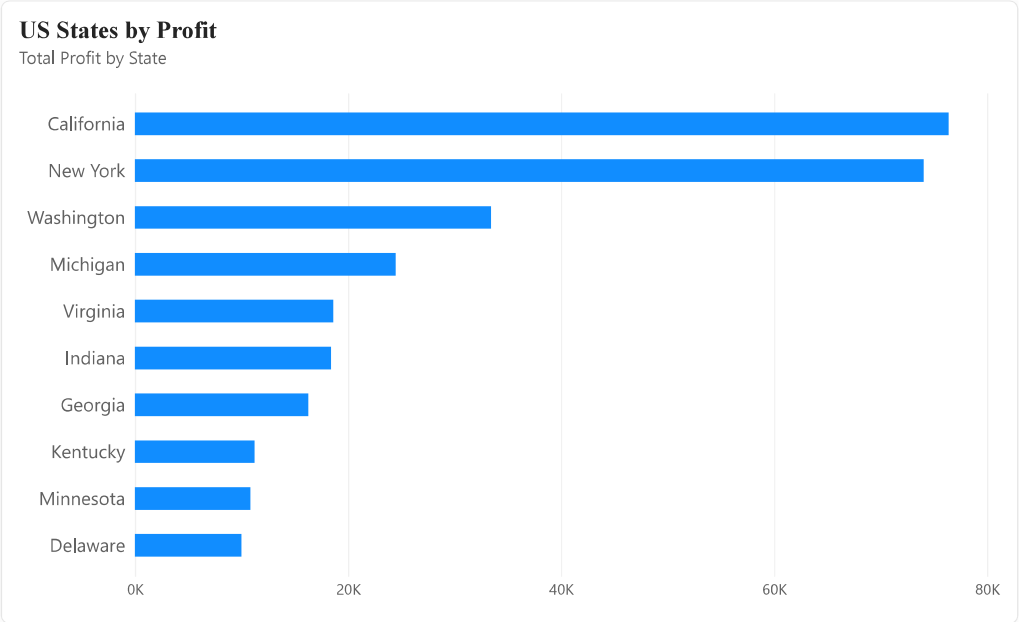
Top 10 Loss Making Product

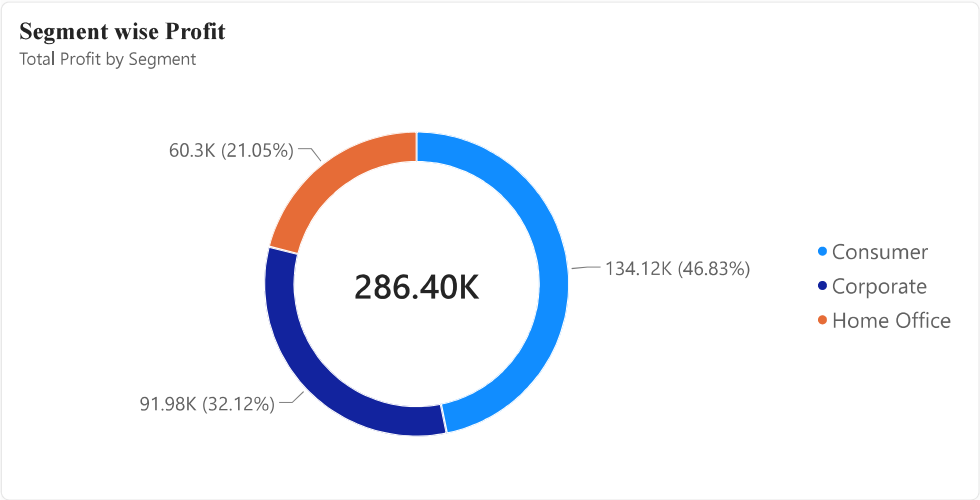
Product Name	Total Sales	Total Profit	Average of Discount
Martin Yale Chadless Opener Electric Letter Opener	16,656.20	-1,299.18	0.10
Lexmark MX611dhe Monochrome Laser Printer	16,829.90	-4,589.97	0.40
GBC DocuBind P400 Electric Binding System	17,965.07	-1,878.17	0.45
Cubify CubeX 3D Printer Triple Head Print	7,999.98	-3,839.99	0.50
Cubify CubeX 3D Printer Double Head Print	11,099.96	-8,879.97	0.53
Cisco TelePresence System EX90 Videoconferencing Unit	22,638.48	-1,811.08	0.50
Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	9,917.64	-2,876.12	0.28
Bush Advantage Collection Racetrack Conference Table	9,544.73	-1,934.40	0.35
BoxOffice By Design Rectangular and Half-Moon Meeting Room Tables	1,706.25	-1,148.44	0.48
Balt Solid Wood Round Tables	6,518.75	-1,201.06	0.20

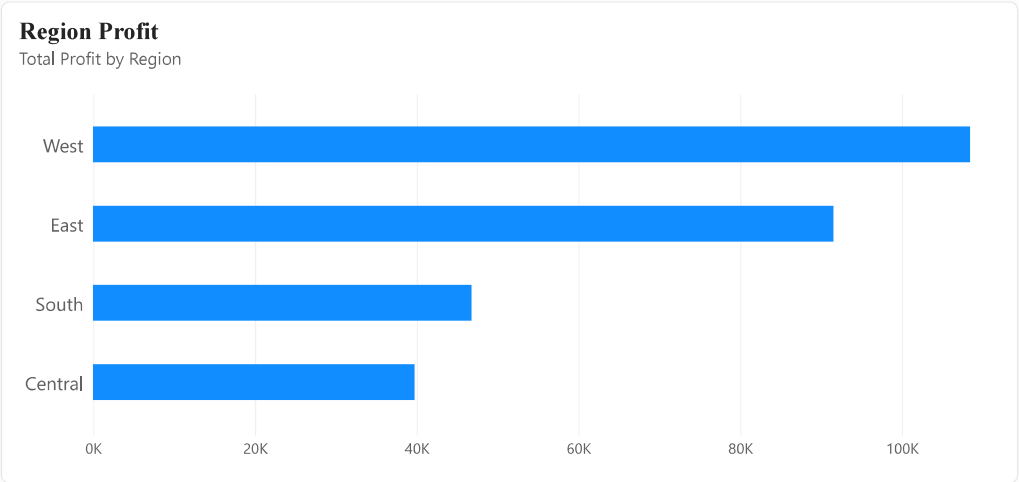
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Region & Customer Analysis

Insight:

West is the most profitable region and Central is the lowest (Central gives the highest average discount, 24% vs 11% in West). California and New York earn the most profit. Texas, Ohio, Pennsylvania and Illinois are net loss-making states because of heavy discounts, so they do not appear in the top 10 chart. Consumer gives the highest profit (134K, 46.8%) but has the lowest margin (11.5%).







Data drive analytic/ consulting capstone project Trend & Recommendations

Insights & Recommendations:

1. Sales show seasonality: Q4 is the peak quarter.
2. Discounts above 20% cause losses: these rows are only 14% of orders but lose 135K. At 40%+ discount, every order is loss-making. Tables and Bookcases lose money mainly due to high discounts.
3. Recommendation: Cap discounts at 20% (management approval above it), and focus growth on Technology and Office Supplies (about 17% margin each).

